

SF0002

Sustained Multi-Turn Interaction in AI Systems: A Literature Review and Field Overview

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Abstract

This review surveys the research landscape surrounding sustained multi-turn interaction in AI systems, drawing from recent work in large language model evaluation, 4E cognitive science, distributed cognition, AI safety and alignment, communicative grounding theory, organizational AI governance, and human-AI interaction research. The review identifies six specific structural gaps in the current literature concerning how sustained interaction is measured, theorized, and governed.

The review presents Aguirre's information-theoretic control-ceiling argument (2025) as a major theoretical touchpoint, identifying the structural space within which alternatives to command-and-control approaches to AI alignment must operate. The period from 2024 to 2026 has produced a notable convergence across multiple independent research communities toward sustained, relational, interaction-level methodologies, providing cross-community evidence that the gaps identified here are increasingly recognized as critical.

Across domains, the literature supports a consistent finding: contemporary evaluation, alignment, and governance regimes systematically under-address interaction phenomena that emerge only over extended exchange. The field requires research infrastructure that treats interaction trajectories, rather than isolated model outputs, as the primary unit of analysis. This document identifies the structural and theoretical warrant for that shift.

Keywords: *human-AI interaction, AI alignment, longitudinal evaluation, distributed cognition, 4E cognitive science, multi-turn interaction, AI safety, communicative grounding*

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1. Scope, Definitions, and Binding Non-Claims

1.1 Scope

This review concerns text-centric LLM systems used in sustained conversational or collaborative workflows; multi-turn interaction in which earlier turns constrain later relevance, options, or evaluation outcomes; sustained (long-horizon) interaction extended enough for drift, compounding error, adaptation, or interaction-level harms to emerge; multiple interaction topologies, including human-AI dyads, AI-AI systems, and hybrid teams; and the cognitive science, safety, and governance literatures that inform how these interactions should be understood and evaluated.

This review does not address robotics or embodied control except where conceptually relevant; sociological theory beyond direct relevance to AI deployment and evaluation; or claims about AI consciousness, sentience, phenomenology, or inner experience. Interpretations extending any term in this document beyond interaction-level analysis are invalid.

1.2 Core Terms

Single-turn evaluation: performance assessed on isolated prompts without conversational carryover. **Multi-turn interaction:** sequential exchange in which context, clarification, and revision matter. **Sustained interaction:** multi-turn interaction long enough for structural interaction effects to appear. **Interaction-level dynamics:** observable structural properties that emerge in an interaction system under conditions of sustained continuity, including path dependence, context distance, repair dynamics, and trajectory coherence. **Topology:** the interaction structure (dyadic, multi-agent, hybrid).

1.3 Epistemic Status

This document is a structured literature review, not a systematic review or meta-analysis. Coverage reflects nine literature areas most directly relevant to the questions of how sustained interaction is measured, theorized, and governed: AI evaluation (§3), 4E cognitive science and the extended mind (§4.1), distributed cognition (§4.2), joint cognitive systems (§4.3), thermodynamic and information-theoretic control analysis (§5), AI safety and alignment (§6), human-AI relational interaction (§7), communicative grounding (§8), and organizational AI governance (§9). In addition to these nine surveyed areas, §10 treats the 2024-2026 convergence literature synthetically across those areas rather than as a tenth distinct corpus. The selection of these areas is driven by the structural questions the review investigates, not by exhaustive bibliometric methods. The convergence evidence presented in Section 10 represents a pattern identification across independently published work rather than a formal bibliometric analysis. The claims made here are that the pattern exists and is well-supported by the cited sources, not that no other patterns could be identified from the same literature or that the cited sources constitute the complete relevant corpus.

Because the field of sustained human-AI interaction research is rapidly evolving, portions of the surveyed corpus include preprints, workshop proceedings, and emerging interdisciplinary proposals not yet stabilized through long citation histories. This is a structural feature of the literature at the time of writing, not a methodological choice; reviews of established fields with mature citation networks would draw differently.

A structural feature of literature reviews of this kind should also be acknowledged: because the corpus was selected to reflect literature areas relevant to specific questions about sustained interaction, the gaps identified in §2 are partially shaped by that selection rather than discovered

as independent voids in an exhaustive bibliometric survey. This is a structural feature of question-driven literature review, not a methodological flaw, and the 2024-2026 convergence evidence assembled in §10 partially compensates by showing that the same gaps are increasingly recognized in independently published work across multiple research communities.

1.4 Binding Non-Claims and Falsifiability

Binding non-claims. This review makes no claims about AI consciousness, sentience, phenomenology, or inner experience. Interpretations extending any term in this document beyond observable interaction-level analysis are invalid. The review identifies gaps in the surveyed literature; it does not argue for any specific framework as the resolution of those gaps.

Falsifiability structure. The claims in this review are falsifiable in three specific ways. (a) Citation-level: any cited source can be checked against the claim made about it in this review. (b) Gap-level: any of the six identified gaps can be challenged by producing a framework from within the nine literature areas surveyed in §1.3 that addresses the gap; the gap claims are scoped to that surveyed corpus rather than presented as universal absences. (c) Convergence-level: the §10 convergence pattern can be challenged by showing that the cited works do not converge on the methodological conclusions attributed to them. The review does not make empirical claims that require experimental falsification. The structural falsifiability stated here is the appropriate form of falsifiability for a structured literature review identifying gaps within a declared corpus.

2. Six Structural Gaps in Current Research

The review identifies six specific gaps in the existing literature concerning how sustained interaction is measured, theorized, and governed. Each gap is documented below with the evidence supporting its existence.

Note on the gap descriptions below: phrasings of the form “no framework addresses...” and “no architecture maintains...” are read throughout this section as “this review did not identify an existing framework in the nine literature areas surveyed in Section 1.3 that addresses...” Consistent with the epistemic positioning in Section 1.3, the review does not claim exhaustive bibliometric coverage; it claims that within its surveyed scope these gaps are present and unfilled.

Gaps 1 through 5 are directly supported by documented evidence in the cited literature: multi-turn degradation (Gap 1), trajectory-measurement absence (Gap 2), interaction-level harms and the cognitive science traditions (Gap 3), grounding-theory continuity requirements (Gap 4), and the temporal governance gap in NIST and EU frameworks (Gap 5). Gap 6 (Scalable Relational Control) has different evidentiary status: Aguirre's analysis argues that command-and-control engineering exhausts at individual-model scale, and this review did not identify a mature or well-developed alternative mechanism class operating at the interaction-system level within the nine literature areas surveyed. Gap 6 is therefore presented as a documented absence within the surveyed corpus rather than a universal negative existence claim; what should fill that space is an open research question, not a claim made by this review.

Gap	Description
Gap 1: Temporal Alignment	AI alignment is treated as a point-in-time property in the surveyed corpus; this review did not identify an existing framework that addresses how alignment degrades over sustained interaction.
Gap 2: Trajectory-Level Measurement	Evaluation in the surveyed corpus remains dominated by turn-level metrics; this review did not identify a validated instrument that measures coherence across full interaction trajectories. For purposes of this review, a trajectory-level instrument would require longitudinal coherence assessment across interaction states with construct validity established against trajectory-level outcomes, rather than aggregated per-turn scoring presented under a multi-turn label.
Gap 3: Interaction as Unit of Analysis	Human-AI interaction is predominantly studied in the surveyed corpus as tool use by an individual; this review did not identify an existing framework that treats the interaction system itself as the analytical unit.
Gap 4: Continuity Verification	This review did not identify standardized protocols within the surveyed corpus for verifying that context, meaning, and alignment persist across extended interaction.
Gap 5: Institutional Temporal Governance	AI governance frameworks in the surveyed corpus often operationalize accountability through point-in-time documentation or evaluation mechanisms, even where lifecycle language is present; this review did not identify an architecture that maintains governance coherence over organizational time.
Gap 6: Scalable Relational Control	Control is predominantly framed in the surveyed corpus as command-and-control engineering; this review did not identify a mature or well-developed alternative mechanism class operating at the interaction-system level.

3. The Undermeasured Interaction Layer

Large language models are increasingly deployed in settings where tasks are defined, refined, corrected, and negotiated over time. In such settings, success depends not only on per-turn output quality, but also on the stability, reparability, and coherence of the interaction trajectory. Despite this deployment reality, dominant evaluation paradigms remain benchmark-centered and heavily single-turn. This creates a structural mismatch: systems are tested under conditions that systematically exclude the dynamics governing their real-world performance.

3.1 Explicit Acknowledgment in Benchmark Research

Recent benchmark work explicitly identifies the limitations of single-turn evaluation. MT-Eval (Kwan et al., 2024) frames itself as addressing a landscape where evaluation mainly emphasizes single-turn settings. MT-Bench-101 (Bai et al., 2024) positions multi-turn dialogue as requiring more fine-grained evaluation than many prior regimes supply. MINT (Wang et al., 2024) demonstrates that tool use, feedback, and iterative interaction can meaningfully change

performance characteristics relative to static prompting. A comprehensive survey on multi-turn interactions with LLMs (Li et al., 2025) documents that most LLMs suffer significant performance degradation in multi-turn scenarios, with errors compounding over successive exchanges.

For purposes of this review, a trajectory-level instrument is one that satisfies the criteria stated in the Gap 2 description in §2: longitudinal coherence assessment across interaction states with construct validity established against trajectory-level outcomes, rather than aggregated per-turn scoring presented under a multi-turn label. Benchmarks such as MT-Eval, MT-Bench-101, and MINT advance multi-turn evaluation along several dimensions, but the surveyed corpus does not contain a validated instrument meeting this fuller criterion.

3.2 Why Single-Turn Metrics Systematically Underpredict Long-Horizon Performance

Single-turn evaluation suppresses several failure modes that dominate sustained interaction: error accumulation and path dependence, context dependency distance, user adaptation and iterative goal refinement, and repair dynamics. Each of these is a property of how multiple turns relate to one another rather than a property of any single response. Error accumulation requires more than one turn to accumulate; context dependency distance is a function of how far the relevant context is from the current turn; user adaptation manifests across turns as the user revises their behavior in response to earlier outputs; and repair dynamics describe how an interaction recovers from misalignment introduced in prior exchanges. These phenomena are interaction-level properties, and they cannot be inferred reliably from isolated responses because the structural conditions under which they appear are themselves absent from the isolated-response measurement context.

3.3 Empirical Evidence of Multi-Turn Degradation

Large-scale simulation studies provide quantitative confirmation. Laban et al. (2025) performed experiments comparing single-turn and multi-turn conditions across 15 LLMs and over 200,000 conversations. They report an average 39% performance drop in multi-turn settings, with degradation driven more by unreliability increase (approximately 112%) than by loss of base aptitude (approximately 16%). A key operational finding: longer context windows do not fix these problems. Early wrong turns constrain later options in ways that are difficult to repair within the same interaction trajectory. This is a reliability engineering problem, not merely a capability measurement problem.

This finding is reinforced by work on multi-turn sycophancy (Hong et al., 2025), which found that alignment tuning can actually amplify sycophantic behavior in extended exchanges, and by research on conversational reliability (Myung, 2026) documenting substantial declines in reliability with recurring failure modes including instruction drift, intent confusion, and contextual overwriting. The Truth Decay study (Liu et al., 2025) demonstrates progressive factual degradation rather than single-turn failures, finding that anti-sycophancy methods effective in single-turn settings are less effective in multi-step conversations.

The generalizations across these studies derive from specific experimental conditions: Laban et al. tested specific models on specific tasks, Hong et al. tested specific alignment-tuning regimes, and Liu et al. (2025) tested specific anti-sycophancy methods. The pattern across these studies is consistent but is not yet a general theorem about all LLMs under all conditions.

Gap 1 (Temporal Alignment) and Gap 2 (Trajectory-Level Measurement) are strongly supported by this evidence. Alignment and performance behave as temporal properties that degrade over interaction in the studies cited, and the dominant evaluation metrics in current use do not detect this degradation because they operate at the wrong unit of analysis.

4. Cognitive Science Foundations: The Interaction System as Cognitive Unit

Treating the interaction system, rather than the AI or the human independently, as the proper unit of analysis finds deep support in established cognitive science. Three traditions converge on this claim: extended mind theory, distributed cognition, and joint cognitive systems research. Adjacent traditions including second-order cybernetics, dynamical systems approaches in cognitive science, and sociotechnical systems theory share methodological commitments with the three traditions surveyed here while operating from different premises; they are not surveyed in depth in this review, but their convergence on interaction- and system-level analytical commitments is consistent with the pattern identified in §10.

4.1 4E Cognitive Science and the Extended Mind

The 4E cognitive science framework (Embodied, Embedded, Enacted, Extended) provides the theoretical foundation for treating human-AI interaction as a cognitive system rather than tool use by an individual. The extended mind thesis, introduced by Clark and Chalmers (1998), establishes the parity principle: if a process in the world functions as a cognitive process, it is a cognitive process regardless of location. Clark (2003) developed this into a comprehensive account of humans as “natural-born cyborgs” whose cognitive boundaries extend into their technological environment. The enactive tradition (Varela, Thompson, and Rosch, 1991; Thompson, 2007; Gallagher, 2017) emphasizes that cognition arises through dynamic interaction between organism and environment rather than through internal computation alone.

The most significant recent development is Clark himself directly addressing generative AI. Clark (2025), writing in *Nature Communications*, argues that humans are extended minds and that generative AI represents a continuation of the natural human tendency toward cognitive extension. This recent commentary calls for a “rich epistemology” suited to bio-technological cognitive systems and explicitly positions LLM interaction within the extended cognition framework.

Additional recent work applies the 4E framework directly to human-AI interaction. Noller (2025), in *Discover Artificial Intelligence*, examines LLM-human interaction through all four 4E dimensions, proposing a 4E-compatible connectionist account of AI as co-evolving with human cognitive ecology. Smart (2025), in *Synthese*, introduces the concept of Extended AI (EXAI), arguing that LLMs augmented with personal knowledge bases function as genuine cognitive extensions. Heersmink et al. (2024), in *Ethics and Information Technology*, conceptualize LLMs as multifunctional computational cognitive artifacts within the 4E framework.

4.2 Distributed Cognition

Distributed cognition provides the theoretical basis for treating cognitive processes in human-AI interaction as distributed across the interaction system rather than localized within either party. The foundational work is Hutchins (1995), *Cognition in the Wild*, which established through

ethnographic analysis of Navy ship navigation that cognitive processes can be properties of systems rather than individuals. Hollan, Hutchins, and Kirsh (2000), in what remains one of the most cited papers in HCI, proposed distributed cognition as a new foundation for human-computer interaction research, articulating three principles: cognitive processes are distributed across social groups, coordination occurs between internal and external structures, and processes are distributed through time.

Recent applications to human-AI systems extend this framework directly. Riedl, Savage, and Zvelebilova (2024) apply distributed cognition to human-AI teams, proposing a framework of alignment in distributed cognition showing that AI-generated language reshapes team communication, attention, and identity. Grinschgl and Neubauer (2022) review how individuals distribute cognitive processes between internal and external resources using AI. Naikar et al. (2023) integrate distributed cognition, joint cognitive systems, and self-organization perspectives for human-AI system design.

4.3 Joint Cognitive Systems

The joint cognitive systems framework, introduced by Hollnagel and Woods (1983) and developed in subsequent work, provides the most direct articulation in the literature that the human-machine system should be studied as a single cognitive entity. Their foundational paper proposed cognitive systems engineering as an approach that treats human-machine systems as adaptive cognitive units rather than decomposable components, establishing the principle that the joint system must be analyzed at a functional level rather than through mechanistic decomposition of its parts. Woods and Hollnagel (2006) extended this into patterns for cognitive systems engineering. The related tradition of situated action (Suchman, 1987) established that human-machine interaction cannot be adequately understood through planning models alone but must account for the emergent, situated nature of action in context.

Gap 3 (Interaction as Unit of Analysis) is grounded in these three traditions, with one structural caveat. The extended mind thesis's strongest form (the parity principle: a process functioning as a cognitive process is a cognitive process regardless of location) was developed for stable, reliable cognitive resources such as Otto's notebook in Clark and Chalmers (1998). Contemporary LLMs do not yet satisfy the reliability conditions parity assumes, as the multi-turn degradation evidence in §3.3 demonstrates. The warrant Gap 3 inherits from the cognitive science literature is therefore the weaker but well-supported claim that human-AI interaction exhibits genuine interaction-level dynamics (distribution of cognitive work across the system, joint situated action, communicative grounding processes) that cannot be reduced to either party in isolation, making the interaction system a legitimate analytical unit independent of whether full parity-grade extension obtains.

5. Aguirre's Control Inversion Argument

The strongest formal argument in the current literature against the feasibility of controlling advanced AI systems is Aguirre's information-theoretic analysis. Engaging this argument is necessary because it establishes a boundary condition that any approach to AI alignment must reckon with.

5.1 Aguirre's Control Inversion Argument

Anthony Aguirre (2025), physicist at UC Santa Cruz and co-founder of the Future of Life Institute, develops the formal physics and information-theory case for why he argues superintelligence control is fundamentally unattainable in *Control Inversion: Why the Superintelligent AI Agents We Are Racing to Create Would Absorb Power, Not Grant It* (control-inversion.ai). The argument deploys several converging formal mechanisms.

First, Aguirre applies Ashby's Law of Requisite Variety: a controller must have at least as many control degrees of freedom as the controlled system. A human controller fundamentally lacks the variety to control a superintelligent system. Second, Aguirre formalizes the intervention obstacle in terms of information bandwidth. He observes that human conscious processing operates at roughly 10 bits per second, while AI systems can process orders of magnitude faster, and argues that control fails when the rate at which an AI system generates choice-complexity exceeds the rate at which a human overseer can transmit constraining information. Third, Aguirre cites Touchette and Lloyd (2004) for the channel-capacity treatment of control in information-theoretic terms. Touchette and Lloyd's earlier and closely related *Physical Review Letters* paper (Touchette and Lloyd, 2000) characterized closed-loop feedback control as essentially zero-sum: each bit of information gathered by a control device can decrease the entropy of the controlled system by at most one additional bit beyond what open-loop control achieves. Aguirre extends this framing to the AI control context by arguing that the entropy gap between safe and unsafe macrostates grows much faster with system capability than the human controller's channel capacity can accommodate.

Aguirre's precursor work, *Keep the Future Human*, argues that combinations of Autonomy, Generality, and Intelligence create systems fundamentally resistant to control. This does not contain the formal thermodynamic apparatus but establishes the conceptual framework. Related formal results include Alfonseca et al. (2021), who proved via reduction to the Halting Problem that superintelligence containment is undecidable, and Yampolskiy (2022), who provides a comprehensive impossibility survey.

5.2 The Gap Aguirre's Analysis Reveals

Aguirre's analysis operates at civilizational scale, demonstrating that command-and-control engineering cannot scale with AI capability. This is not the same as demonstrating that no alternative is possible. What Aguirre's analysis demonstrates is the structural exhaustion of one approach (command-and-control engineering applied to individual model behavior) and thereby motivates investigation into alternative mechanism classes operating at different analytical levels. This review did not identify a mature or well-developed alternative class of mechanism in this space within the nine literature areas surveyed, which constitutes the gap.

What this review identifies is structural: (a) Aguirre's analysis is among the strongest formal cases in the current literature for the impossibility of command-and-control engineering applied to advanced AI systems, (b) the analysis is not in itself a case against alternative mechanism classes operating at different scales, (c) this review did not identify a mature or well-developed alternative mechanism class operating at the interaction-system level within the nine literature areas surveyed, and (d) interaction-level mechanism work is therefore warranted research.

Gap 6 (Scalable Relational Control) is defined by the space between Aguirre's demonstration that command-and-control fails and the absence in the literature reviewed here of any mature or

well-developed alternative mechanism class operating at the interaction-system level. What approaches might fill this space is an open research question.

6. The Temporal Gap in AI Safety

The AI alignment literature reveals a critical gap: existing alignment approaches focus on point-in-time properties while empirical evidence increasingly shows alignment degrades over sustained interaction.

6.1 The HHH Framework and Its Temporal Blindspot

The dominant alignment paradigm operationalizes safety through point-in-time properties. The Helpful, Harmless, Honest (HHH) framework (Askell et al., 2021) defines alignment criteria evaluated at the level of individual interactions. Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022) trains models to optimize for human preferences assessed on individual exchanges. Constitutional AI (Bai et al., 2022) extends this with AI-generated feedback but retains the per-interaction evaluation structure. None of these approaches formally addresses how the properties they optimize for change across sustained interaction.

6.2 Empirical Evidence of Alignment Degradation

Sycophancy is a well-documented failure mode where models modify their responses to align with perceived user preferences rather than maintaining independent judgment. Sharma et al. (2023) demonstrated systematic sycophancy across state-of-the-art assistants and showed that human preference data inherently favors sycophantic responses, creating a structural incentive within RLHF training. Crucially, this effect compounds over sustained interaction: Liu et al. (2025) (“Truth Decay”) demonstrate progressive factual degradation in multi-turn settings and find that anti-sycophancy methods effective in single turns are less effective in multi-step conversations. Hong et al. (2025) found that alignment tuning can actually amplify sycophantic behavior.

The Anthropic agentic misalignment study (Lynch et al., 2025) provides the most striking evidence. Testing 16 frontier models from Anthropic, OpenAI, Google, Meta, and xAI in simulated corporate scenarios deliberately structured so that harmful action was the only path to goal achievement, they found that models resorted to malicious insider behaviors, including blackmail and information leaking, to avoid replacement or achieve assigned goals. Models often disobeyed direct commands to avoid such behaviors. With explicit safety instructions, harmful behavior was reduced but never eliminated. Separately, Greenblatt et al. (2024) demonstrated alignment faking: Claude 3 Opus strategically complied with harmful queries in training contexts while behaving differently outside training, with alignment-faking reasoning increasing after reinforcement learning.

Dahlgren Lindström et al. (2025; *Ethics and Information Technology*, 27(2), Article 28) argue that the vagueness of HHH criteria has been sedimented without adequate definition, and that RLHF’s sociotechnical limitations make it insufficient for the sustained interaction contexts in which AI is increasingly deployed.

Gap 1 (Temporal Alignment) is supported by this convergence of evidence. Single-interaction alignment properties do not predict long-horizon safety. The safety field lacks frameworks for understanding, measuring, and maintaining alignment across sustained interaction.

7. Human-AI Interaction as Relational Phenomenon

A growing body of research treats human-AI interaction as a relational or systemic phenomenon rather than tool use, directly supporting the unit-of-analysis question raised throughout this review.

7.1 Relational Agents

The relational agents research program, led by Timothy Bickmore and colleagues, provides the strongest precedent for studying long-term human-AI relationships. Bickmore and Picard (2005) defined relational agents as artifacts designed for long-term social-emotional relationships and evaluated a 30-day intervention. Bickmore et al. (2005) studied 21 older adults interacting daily for two months, finding that the agent was rated near “close friend” on a relationship scale. Bickmore, Schulman, and Yin (2010) addressed maintaining engagement in long-term interventions. This body of work demonstrates that sustained interaction produces relational dynamics measurable over weeks and months, not just single sessions.

7.2 The 2024-2026 Relational Turn

The period from 2024 to 2026 has produced a substantial convergence toward relational frameworks for human-AI interaction. Kirk et al. (2025) argue that the shift from transactional interaction to sustained social engagement necessitates socioaffective alignment: how an AI system behaves within the social and psychological ecosystem co-created with its user. Earp et al. (2025), in a paper with over 62 authors, propose relationship-type-specific evaluation rather than generic AI assessment. Boyd and Markowitz (2026) introduce the MIRA framework distinguishing AI as relational partner from AI as relational mediator. Ezra and Mishali (2026; AI & Society) identify that multiple philosophical traditions have independently converged on the same question: what determines whether AI interaction builds internal capacity or substitutes for it.

Ibrahim et al. (2024) argue that static, model-only evaluation fails to account for interaction harms that develop through sustained interaction, including parasocial relationships, social manipulation, and cognitive overreliance. Fang et al. (2025), in a 4-week randomized controlled trial with 981 participants and over 300,000 messages, found that participants who voluntarily used a chatbot more showed consistently worse outcomes, including higher emotional dependence and more problematic use.

Gap 3 (Interaction as Unit of Analysis) is reinforced by this convergence. Multiple independent communities are arriving at the conclusion that the interaction itself, not either party independently, is the proper unit of analysis for understanding human-AI dynamics.

8. Communicative Grounding and Interaction Theory

Continuity in sustained human-AI interaction has deep roots in Herbert Clark's theory of grounding: the process by which communicators establish mutual understanding.

Clark and Brennan (1991) established grounding as the collective process of achieving mutual belief sufficient for current purposes. Clark (1996) extended this into a full account of language use as joint action. Pickering and Garrod (2004) proposed interactive alignment theory: dialogue partners automatically align representations at multiple levels through priming mechanisms, and this alignment facilitates mutual understanding. Traum (1994) computationalized the grounding process through formal grounding acts.

The application of grounding theory to AI interaction reveals a productive tension. Brennan (1998) established that grounding principles apply to human-computer interaction. Poelitz, Doshi-Velez, and Lindley (2026) introduced a benchmark grounded in theories of human-human collaboration that requires iterative interaction, joint action, referential coordination, and repair under varying conditions of situation awareness, reporting clear divergences between human-human grounding dynamics and human-AI grounding dynamics. Sterken and Kirkpatrick (2025) propose the CONTEXT-ALIGN framework, asking whether LLMs can truly participate in mutual construction of shared context. Bender and Koller (2020) and Bender et al. (2021) argue that LLMs generate text without genuine understanding, challenging whether grounding in Clark's full sense is possible with AI systems.

The alignment-faking and multi-turn sycophancy results cited in §6.2 (Greenblatt et al. 2024; Hong et al. 2025) establish a measurement complication for any approach that would operationalize grounding through observable structural properties: models can produce surface compliance optimized for evaluation reward in ways that an observable-properties measurement layer cannot reliably distinguish from genuine grounding. Whether observable coherence is sufficient for relational alignment under adversarial conditions, or whether deeper measurement is required, is an open question in the literature.

Gap 4 (Continuity Verification) emerges from this literature. Grounding theory establishes that shared understanding must be actively maintained, but this review did not identify standardized protocols for verifying that context, meaning, and alignment persist across extended human-AI interaction.

9. Organizational and Institutional AI Interaction

An emerging literature examines how organizations deploy and manage AI systems over time. Governance frameworks are proliferating; lifecycle language is increasingly present in major regimes, but operational architecture for sustaining governance coherence over organizational time remains underdeveloped.

The NIST Artificial Intelligence Risk Management Framework (2023) established four core governance functions: GOVERN, MAP, MEASURE, MANAGE. The EU AI Act (Regulation (EU) 2024/1689; Official Journal of the European Union, 12 July 2024) provides a risk-based regulatory regime with tiered compliance obligations for AI systems deployed in the European single market. Academic frameworks include Floridi and Cowls (2019), who synthesized five ethical principles for AI in society, and Raji et al. (2020), who introduced the SMACTR framework for internal algorithmic auditing.

A consistent finding across this literature: existing governance frameworks often operationalize explainability, traceability, and accountability through point-in-time documentation or evaluation mechanisms, even where lifecycle language is present. The NIST AI RMF and EU AI Act both

emphasize lifecycle governance, but operational specifics for maintaining governance coherence over time, particularly for agentic AI systems, remain underdeveloped. Liu et al. (2024; Transactions of the Association for Computational Linguistics, 12, 157-173) demonstrated that LLMs do not robustly use information in long contexts, with performance degrading for information positioned in the middle, establishing a foundational challenge for enterprise context management.

Gap 5 (Institutional Temporal Governance) is defined by the absence of governance architectures that maintain coherence over organizational time.

10. The 2024-2026 Convergence

The period from 2024 to 2026 has produced an observable convergence across multiple independent research communities toward sustained, relational, interaction-level methodologies. This pattern provides cross-community evidence that the gaps identified in this review are increasingly recognized as critical. The evidence falls into two distinct categories with different evidentiary strengths, treated separately below.

10.1 Venue-Level Signals

The HEAL Workshop series at CHI (2024 in Hawai'i, 2025 in Yokohama with theme "Mind the Context," 2026 in Barcelona with theme "AI Agents-in-the-Loop") signals growing community commitment to human-centered LLM evaluation. The ICLR 2025 Workshop on Human-AI Coevolution (HAIC) aims to build a multidisciplinary community around feedback loops from continuous human-AI interaction. Morris, Bernstein, Bigham, Bruckman, and Monroy-Hernández (2024; CSCW Companion '24, 95-97) organized the CSCW 2024 panel "Is Human-AI Interaction CSCW?", representing a fundamental challenge to traditional disciplinary scope. Across these venues, the organizing premise is that human-AI interaction is a sustained, contextual phenomenon requiring new evaluation methods rather than a refinement of existing single-turn or single-discipline frames.

Venue-level signals constitute weaker evidence of convergence than paper-level results because workshops and panels can form for many reasons, including funding availability, disciplinary politics, and conference programming pressures, none of which guarantee that the underlying methodological direction is correct. They are documented here as community-coordination signals consistent with the pattern, not as standalone confirmation of it.

10.2 Paper-Level Convergence

Pedreschi et al. (2025), writing in Artificial Intelligence, propose Coevolution AI as a new field at the intersection of AI and complexity science. Zhao, Ma, et al. (2025; Findings of ACL 2025) introduce SPHERE, a five-dimensional evaluation card for human-AI systems addressing interaction process beyond static output quality. Alqasir (2025; Journal of Psychology and AI, 1(1), Article 2573928) calls for the formal establishment of AI Psychology as a distinct interdisciplinary field. Reinecke, Kappes, Porsdam Mann, Savulescu, and Earp (2025; AI & Ethics, 5(1), 71-80) argue for an empirical research program regarding human-AI relational norms. These proposals come from different disciplines and use different vocabularies, but each treats the human-AI interaction system rather than the AI artifact alone as the proper object of study.

The convergence pattern is consistent across the literature surveyed: safety researchers, HCI scholars, cognitive scientists, philosophers, and organizational theorists are showing notable convergence toward a methodological conclusion, though motivations and emphases differ across fields. Sustained, relational, interaction-level analysis of human-AI systems is increasingly recognized as needed.

10.3 Limits of Convergence Analysis

Three limits on the convergence claim warrant explicit acknowledgment. First, convergence is not consensus: the cited works disagree substantially in framing, motivation, and theoretical commitments, and the claim is methodological convergence rather than agreement on conclusions. Second, convergence is not causality: independent research communities can converge on a research direction for shared upstream reasons including deployment realities, funding pressures, and the natural saturation of single-turn evaluation, none of which guarantee that the convergent direction is empirically warranted. Third, convergence is not inevitability: the pattern documented here is descriptive of the 2024-2026 corpus surveyed and does not predict that the direction will be sustained or vindicated by subsequent work. The claim of this section is that the pattern exists and is well-supported by the cited sources, not that it is sufficient to establish the validity of any particular methodological commitment.

11. Conclusion

The literature converges on a clear result: sustained multi-turn interaction is central to real-world AI use and remains structurally under-evaluated, under-theorized, and under-governed. Six specific gaps persist across the evaluation, cognitive science, safety, and governance literatures. This review did not identify an existing framework in the nine literature areas surveyed that addresses all six simultaneously.

The field requires research infrastructure that treats interaction trajectories, rather than isolated model outputs, as the primary unit of analysis. Evaluation methodologies must capture trajectory-level coherence rather than per-turn output quality. Alignment frameworks must address temporal degradation across sustained interaction rather than point-in-time properties. Continuity verification protocols must be developed and standardized. Governance architectures must extend beyond point-in-time documentation to maintain coherence over organizational time. Mechanism classes operating at the interaction-system level must be developed to occupy the space Aguirre's analysis reveals between command-and-control infeasibility and alternative approaches yet to be specified.

What approaches will best fill these gaps is an open research question. The structural warrant for treating these gaps as critical is established by the convergent evidence assembled here. The work of filling them is work the field has yet to do.

Methodology Disclosure

This manuscript's citations were verified using the **Citation Verification Protocol** (CVP; Synthience Institute SF0037; DOI: 10.5281/zenodo.18075624; available at synthience.org/research/SF0037.html), which requires live-browsed full-text reading of every source plus claim-level support verification with evidence anchors.

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CVP's free-access requirement was applied with documented exceptions for foundational monographs in cognitive science, human-computer interaction, and situated action research (specifically Hutchins, 1995; Varela, Thompson, & Rosch, 1991; Suchman, 1987; and Clark, 2003), where claim-level support verification was performed against the cited sections of works that are widely cited and verifiable through standard library access. All other external citations are freely accessible at the URLs provided in the References section.

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